
Algorithm 7 Evolved loss: fit functional shape, then reward calibration

Input Expression t and data (X, y) with n examples.

- 1 $z \leftarrow t(X)$; return $+\infty$ if evaluation fails, lengths disagree, or $n = 0$
- 2 **Accumulate** $\sum_i z_i, \sum_i y_i, \sum_i z_i^2, \sum_i y_i^2, \sum_i z_i y_i$
- 3 $\bar{z} \leftarrow n^{-1} \sum_i z_i; \bar{y} \leftarrow n^{-1} \sum_i y_i$
- 4 $V_z \leftarrow \max(\sum_i z_i^2 - n\bar{z}^2, 0)$; $V_y \leftarrow \max(\sum_i y_i^2 - n\bar{y}^2, 0)$
- 5 $C_{zy} \leftarrow \sum_i z_i y_i - n\bar{z}\bar{y}$
- 6 **Return** $+\infty$ if any centered moment is nonfinite
- 7 $\sigma \leftarrow \sqrt{V_y/n}$ if $V_y > 0$, otherwise $\max(1, |\bar{y}|)$
- 8 $a \leftarrow C_{zy}/V_z$ if $V_z > 0$, otherwise 0; $b \leftarrow \bar{y} - a\bar{z}$
- 9 **Return** $+\infty$ if a or b is nonfinite
- 10 $A \leftarrow 0$; $R \leftarrow 0$ # Second pass: directly evaluate residuals
- 11 **for** $i = 1, \dots, n$ **do**
- 12 $A \leftarrow A + ((az_i + b - y_i)/\sigma)^2$
- 13 $R \leftarrow R + ((z_i - y_i)/\sigma)^2$
- 14 $q \leftarrow R/n$; $h \leftarrow q/(1 + q)$ if q is finite, otherwise 1
- 15 $\ell \leftarrow \sqrt{\text{clip}(A/n, 0, 1)} + h/256$
- 16 **return** $\max(\ell, 0)$ if finite, otherwise $+\infty$

For a nonconstant target, the principal term is the square root of the normalized MSE after the best affine fit $az + b$. The bounded raw-error term favors already calibrated expressions and contributes at most $1/256$. The fitted coefficients are used only to score the tree: this operator does not rewrite it as $at + b$.

The first pass uses plain power sums, with centered variances floored at zero. The second pass evaluates fitted residuals directly (the Julia source uses fused multiply-add) to reduce cancellation near a shape match. Constant predictions use a mean-only fit; constant targets use the magnitude scale $\max(1, |\bar{y}|)$.
